AquaOptima Prototype

Largemouth Bass Aquaculture Optimizer using PSO + EA

The Ultimate Goal

AquaOptima optimizes the management policies for Largemouth Bass aquaculture. The project simulates the lifetime of a fish swarm in a 3D near-realistic pond environment.

Particle Swarm Optimization (PSO) drives individual fish behavior — swimming, foraging, avoiding hazards, and social schooling.

An Evolutionary Algorithm (EA) evolves the pond management configuration — feeding schedules, probiotic treatments, oxygen pumping, and stocking density — to maximize fish yield while minimizing cost.

General Implementation

Inputs

- **MAX_BUDGET**: Maximum spending allowed. Ponds exceeding this are eliminated.
- **AQUACULTURE_DAYS**: Duration of the simulation (each day = 24 timesteps = 24 simulated hours).
- **POND_GENERATIONS**: Number of EA generations per timeline.
- **RUN_TIMELINES**: Number of independent parallel evolutionary runs.
- **INITIAL_POND_COUNT**: Population size (number of pond configurations per generation).
- **POND_WIDTH, POND_HEIGHT, POND_DEPTH**: 3D pond dimensions.

Pipeline

1. For each timeline:
    1. Generate N random pond genotypes.
    2. For each generation:
        1. Simulate all ponds in parallel (all CPU cores).

    2. Evaluate fitness for each pond.
    3. Select next generation using Elitism + Tournament.
  3. The best pond in the **last generation** becomes the Converged Champion.
2. Select the Grand Champion — the best Converged Champion across all timelines.
3. Re-run the Grand Champion with frame recording for 3D visualization.
4. Generate analysis plots from CSV results.

Time Model

- **TimeStep**: 1 simulated hour.
- **Day**: 24 timesteps.
- **Runtime**: AQUACULTURE_DAYS × 24 timesteps.

3D Pond Environment

The simulation runs in a 3D space (width × height × depth). Z=0 is the water surface, Z=POND_DEPTH is the floor.

Static Elements

- **Obstacles**: Static rocks/structures grounded on the floor. Count scales with pond area: NUM_OBSTACLES = sqrt(width × height) × 0.15. Objects can rest on top of obstacles. Fish slide along obstacle surfaces (tangential collision).
- **Underwater Plants**: Dense decorative vegetation covering ~90% of the floor. Plants can also spawn from pollutant transformation.

Dynamic Objects

Object	Behavior
Food Pellet	Drops at surface, sinks slowly (SINK_SPEED=0.3/step). Ages only after reaching floo Expires → Pollutant.

Object	Behavior
Probiotic Pellet	Same sinking behavior as food. Restores immunity directly (no boosting mechanic). E faster → Pollutant.
Oxygen Bubble	Spawns in water column, drifts randomly in 3D. Destroyed by NH3 zones. Also spawn naturally.
Fecal	Dropped by fish based on fullness. Sinks fast (SINK_SPEED_HEAVY=0.6). Stacks w radius. Expires → Pollutant.
Dead Fish	Sinks fast. Generates an NH3 bubble that follows it. Expires → Pollutant.
Pollutant	Exists only on floor. Transforms into one of: NH3 (25%), Disease (10%), Parasite (10 (5%), Plant (10%), Obstacle (25%), or Nothing (15%).

Hazard Zones

Hazard	Shape	Behavior
NH3	3D sphere (radius 6-14)	Drifts randomly. Drains fish oxygen AND HP. Destroys oxygen bubble Expires → Pollutant.
Diseas e	Floor disc (height=12)	Spawns from pollutant. Drains fish immunity. Fish with immunity ≤ 0 b infected. Shrinks over time.
Parasit e	Floor disc (height=12)	Spawns from pollutant. Contact chance per timestep. Parasitized fish multiple debuffs. Shrinks over time.

Pond Genotype (EA Genes)

The EA evolves 10 genes per pond configuration:

Gene	Range	Description
fish_count	10–100	Initial stocking density
food_interval	2–24 hours	Time between food drops
food_quantity	5–20 pellets	Pellets per drop
food_location	0–9 (enum)	Drop zone (Center, TL, TR, BL, BR, TC, BC, LC, RC
probiotic_interval	24–168 hours (step 12)	Time between probiotic drops
probiotic_quantity	1–5 pellets	Pellets per drop
probiotic_location	0–9 (enum)	Drop zone
oxygen_interval	1–24 hours	Time between oxygen pump activations
oxygen_duration	1–4 hours	Pump active duration
oxygen_location	0–9 (enum)	Pump zone

Drop Locations (Top-Down View)

```
TL ---- TC ---- TR
|               |
LC   CENTER   RC
|               |
BL ---- BC ---- BR
```

or

RANDOM

Cost Model

Total Cost = (Runtime/food_interval × food_quantity × $0.05)

+ (Runtime/probiotic_interval × probiotic_quantity × $0.75)

+ (Runtime/oxygen_interval × oxygen_duration × $2.50)

Fish Model

Fixed Traits (per fish, randomized at spawn)

Trait	Range	Purpose
MouthSize	3.0–8.0	Determines max prey size for cannibalism
BodySize	4.0–10.0	Defensive stat — smaller fish are cannibal targets

Dynamic Stats (change each timestep)

Stat	Range	Decay/step	Death Condition
HP	80–100	Conditional	Fish dies if HP ≤ 0
Energy	60–100	0.10/step + movement cost	HP drains if Energy ≤ 0
Fullness	50–100	0.15/step	HP drains if Fullness ≤
Oxygen	80–100	0.10/step (passive regen 0.04 outside NH_3)	Fish dies if Oxygen ≤ 0
Immunity	70–100	Regen 0.02/step (non-infected only)	Infected if ≤ 0 in diseas
Velocity	2.0–6.0	Reduced by low HP/Energy/Fullness/Parasite	—

HP Regeneration

HP slowly regenerates (0.1/step) when ALL conditions met:

- Not infected, not parasitized
- Fullness ≥ 70%, Oxygen ≥ 70%, Energy ≥ 30%, Immunity > 70%

Disease Self-Cure

Infected fish have 8% chance per timestep to self-cure when:

- Fullness ≥ 80%, Oxygen ≥ 80%, Energy ≥ 30%, Immunity ≥ 90%

Fish States

State	Trigger	Effects
Infected	Immunity reaches 0 in disease zone	HP drains 0.6/step. Dead infected fish spawn disease zone
Parasitized	Contact with parasite zone (5% chance/step)	-50% fullness efficiency, +50% fullness/energy drain, HP dr 3 steps, -25% velocity. Fish seeks obstacles to scrub (15% chance per contact).

Cannibalism

Triggers when:

- Fullness < 60% of max (CANNIBAL_FULLNESS_THRESHOLD = 0.6)
- No food within 2× eating range
- Chance scales with hunger: base_chance + (1 - fullness_ratio) × hunger_mult
- Predator's MouthSize must exceed prey's BodySize
- Contact within prey.BodySize × 2.0 distance

Result: Prey instantly dies (no dead body spawned), predator gains fullness.

PSO (Fish Swarm Behavior)

Each timestep, every living fish computes a velocity vector from weighted behavioral components:

new_velocity = inertia × old_velocity + Σ(weight_i × direction_i)

Normalized to the fish's effective speed, then applied as position update.

PSO Vectors

Vector	Weight	Trigger	Direction
cFood	3.0 × (1-fullness_ratio)	Fullness < 100%	Toward nearest food
cProbiotic	1.5 × (1-immunity_ratio)	Immunity < 100%	Toward nearest probiotic
cOxygen	2.0 × (1-oxygen_ratio)	Oxygen < 70%	Toward nearest O2 bubble (with interceptio
cSocial	1.5	Neighbors within 60 units	Toward group center
cSelfish	1.5	Neighbors within 8 units	Away from nearest fish
cNH3	4.0	NH3 zone nearby	Away from NH3 center
cDisease	2.5	Disease zone nearby	Away from disease
cParasite	3.0	Parasite zone nearby	Away from parasite
cObstacle	2.0	Obstacle within 40 units	Away from nearest surface (skipped if para fish seeks obstacles to scrub)

Vector	Weight	Trigger	Direction
cRelief	4.0	Fish has parasite	Toward nearest obstacle/wall surface
cRun	5.0 × (distance factor)	Larger-mouthed fish nearby	Away from predator
cWander	0.75 + 0.3	No social neighbors	Toward pond center + random direction

Obstacle Interaction

- Fish sense obstacles within SENSITIVE_DISTANCE and steer around them
- On collision: tangential sliding (velocity component into surface is zeroed, parallel component kept)
- Parasitized fish are exempt from avoidance — they actively seek obstacles for scrubbing

Spatial Optimization

A 3D spatial hash grid reduces fish-to-fish and fish-to-object lookups from $O(n^2)$ to $O(n \times k)$, enabling simulations with 50+ fish at reasonable speed.

EA (Evolutionary Algorithm)

Selection: Elitism + Tournament

- **Elitism**: Top 20% of ponds copied unchanged to next generation (guarantees best solutions are never lost)
- **Tournament** (K = 40% of population): Pick K random ponds, best one becomes a parent. Repeat for second parent. Child = crossover + mutation.

Crossover

Each gene independently inherited from one parent (50/50 chance per gene).

Mutation

Each gene has 25% chance of being randomized to a new valid value.

Convergence

The **Converged Champion** is the best pond in the **last generation** — representing what the EA converged to, not a historical best that the population may have drifted away from.

Fitness Function

Fitness = 0.55 × Yield + 0.33 × Saving_Rate + 0.12 × Healthiness

Component	Formula	Range	Priority
Yield	alive_count / MAX_FISH_COUNT	0–1	Highest — produce maximum living f
Saving Rate	max(0, budget - cost) / budget	0–1	High — spend as little as possible
Healthiness	avg(normalized stats) of survivors	0–1	Low — survivors should be in good c

If budget is exceeded during simulation, fitness = 0 (eliminated).

Gatekeeper

Before simulation: if per-cycle cost exceeds total budget, pond is rejected without running.

Latest Dead Champion

If no pond in any generation produces survivors, the best result from the last generation is used as a fallback (marked [LATEST DEAD]).

Visualization

A 3D interactive visualization built with Three.js renders the Grand Champion's simulation:

- **Fish**: Ellipsoid bodies with tails, colored by selection state. Status rings (infected=orange, parasitized=purple).
- **Objects**: Food (yellow spheres), Probiotic (green spheres with cross), O2 (blue translucent), Fecal (brown), Dead fish (red with X eyes), Pollutant (orange octahedron), Plants (green blades).
- **Hazards**: NH3 (green translucent spheres), Disease/Parasite (floor cylinders).

- **Obstacles**: Brown boxes grounded on floor.
- **Effects**: Death (red ring + skull), Cannibalism (yellow ring + jaw icon). Both pause with simulation.
- **Controls**: Play/Pause toggle, frame stepping (±1, ±3, ±5), FPS input, progress bar.
- **Fish Panel**: Click any fish to see live-updating traits, vitals (5 stat bars), status, and effective speed.
- **Stats Bar**: Live survival %, average stats, infected/parasitized counts, death count, cannibal count.